INDIAN INSTITUTE OF TECHNOLOGY JODHPUR | B.TECH PROJECT EVALUATION (2026)

Numerical Modelling and Machine Learning Prediction of Single-Particle Crushing Mechanics and Fractal Gradation of Recycled Aggregates

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Background & Engineering Motivation

The Construction & Demolition (C&D) Waste Crisis

- **Global Aggregate Demand:** Over 50 billion tonnes/year of virgin stone & sand mined globally, depleting quarries and riverbeds.
- **C&D Waste Generation:** Exceeds 3.0 billion tonnes/year worldwide; India generates >150 million tonnes/year (CPCB 2024), yet <5% is reused structurally.
- **Circular Economy Materials:** Recycled Concrete (RC), Recycled Mortar (RM), and Recycled Brick (RB) offer sustainable substitutes for highway subgrades & concrete.

Why Recycled Aggregates Crush Prematurely

- **Multi-Phase Heterogeneity:** Unlike virgin basalt/granite, recycled aggregate retains porous adhered mortar & two weak Interfacial Transition Zones (ITZs).
- **Flaw & Water Vulnerability:** High micro-crack density gives 4–10% water absorption (vs. <1.5% virgin rock) and wide strength scatter.
- **Compaction Degradation:** Under vibratory rollers & traffic loads, grains crush into excess fines (<0.075 mm), causing drainage blockages & rutting.

Core Goal: Screen and predict single-particle crushing mechanics & post-breakage gradation non-destructively in <0.1 seconds before road/concrete placement.

Problem Definition & Research Objectives

Evaluation Aspect	Traditional Lab Testing (UTM + Sieving / IS 2386)	Proposed Physics-Informed ML Suite
Test Nature	100% Destructive — specimen lost after single crush	Non-Destructive — zero sample destruction
Testing Speed	16–24 hours per batch (calipers, UTM, mechanical sieving)	< 0.1 seconds per particle (real-time inference)
Occupational Safety	Generates hazardous respirable crystalline silica dust	Completely digital & dust-free screening
Shape Sensitivity	Coarse bulk indices (ACV/LAA) hide 3D shape effects	Captures continuous 3D Sphericity (Φ) & L, I, S axes
Output Completeness	Single aggregate strength or isolated sieve value	Simultaneous Force, Strength, Energy & Fractal Curve

Four Core Project Objectives (Within 4.75–37.5 mm Quasi-Static Scope)

- **1. Physics-Coupled Benchmark Dataset:** Synthesize N = 3,600 calibrated particles (RC, RM, RB) uniting Hiramatsu-Oka, Bagheri, Cai et al. (2024), and Liang et al. (2025) fracture laws.
- **2. Multi-Target ML Benchmarking:** Compare 6 regressors (CatBoost, LightGBM, XGBoost, Random Forest, Physics MLP, Ridge) across Force, Strength, Energy & Fractal Dimension.
- **3. Explainable AI (SHAP) Verification:** Quantify physical feature importance using TreeSHAP to confirm models obey brittle fracture physics rather than overfitting noise.
- **4. Interactive Streamlit Engineering Suite:** Deploy a 4-tab tool (Virtual UTM, Scaling Simulator, Auxiliary CV Photo Sizer, SHAP Engine) validated against IRC:37-2018 guidelines.

Governing Fracture Physics & 3D Morphology

Geometry & Tensile Rupture (Bagheri & Hiramatsu-Oka)

- **Triaxial Dimensions:** Longest (L), Intermediate (I), Shortest (S) axes; 3D Sphericity $\Phi = \sqrt[3]{[(I \cdot S)/L^2]}$
- **Effective Diameter (Eq. 4.2):** $d_o = 0.887 \cdot \sqrt[3]{(L \cdot I \cdot S)}$ [Bagheri et al., 2015]
- **Peak Force–Strength (Eq. 4.3):** $F_f = (\sigma_f \cdot d_o^2) / 0.9$ [Hiramatsu & Oka, 1966]

Weibull Scaling & Fractal Gradation (Liang / Cai / Turcotte)

- **Strength Scaling (Eq. 4.4):** $\sigma f(d_o, \Phi) = [p \cdot \Phi^q] \times [d_o / d_{ref}]^{(-3/m)}$
- **Crushing Energy (Eq. 4.5):** $E_c(d_o, \Phi) = E_o \cdot [d_o / d_{ref}]^{(\alpha_E)} \cdot \Phi^{(\beta_E)}$
- **Fractal Gradation (Eq. 4.6):** $F(d_i) = \min[100, (d_i / d_o)^{(3 - D)} \times 100]$

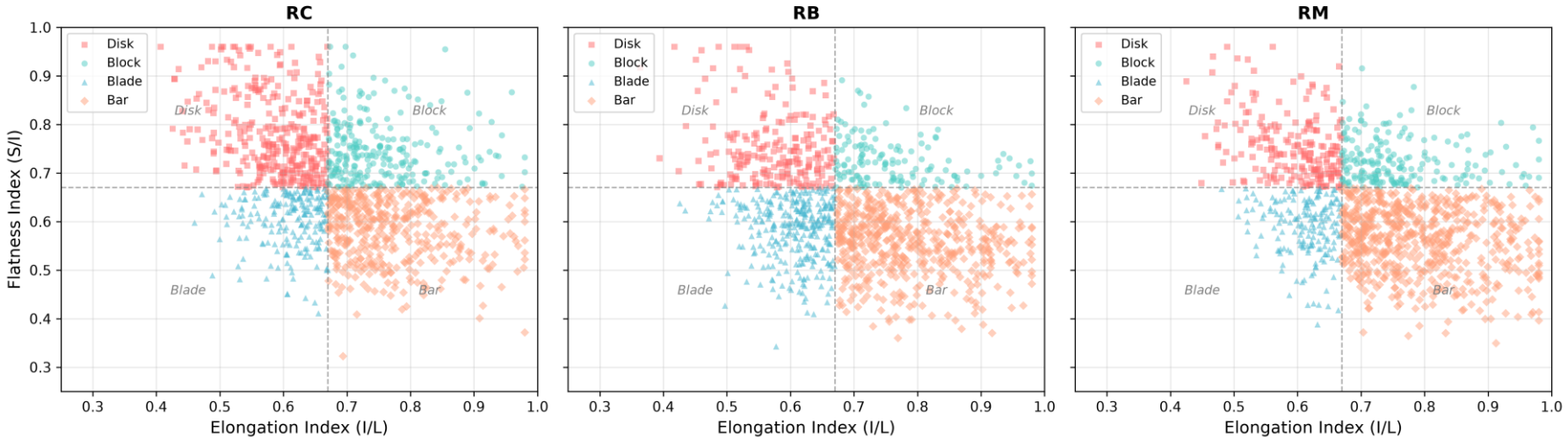


Fig. 4.1: Zingg 3D particle shape classification across Flat (S/I < 0.67), Equant, Blade, and Elongated (I/L < 0.67) morphological domains.

Physics-Coupled Calibration Dataset (N = 3,600)

Material Type	Weibull m	d_ref (mm)	σ_f0 (MPa)	Param p	Param q	E_o (J)
Recycled Concrete (RC)	2.65	25.0	7.15	9.35	2.13	14.2
Recycled Mortar (RM)	2.21	25.0	3.42	4.69	2.21	6.8
Recycled Brick (RB)	1.89	25.0	2.10	3.12	1.95	4.1

Benchmark Dataset Architecture (Table 4.1)

- **Total Samples:** N = 3,600 synthetic particles calibrated strictly to Liang et al. (2025) & Cai et al. (2024) experimental data.
- **Three Sieve Brackets:** 400 particles per class across 4.75–9.5 mm, 9.5–19.0 mm, and 19.0–37.5 mm.
- **Material Ranking:** RC exhibits highest tensile strength (7.15 MPa at 25 mm) due to virgin aggregate cores; RB is weakest (2.10 MPa) due to porous clay.
- **Controlled Dispersion:** Realistic Weibull variability injected via survival probability sampling $P_s = \exp(-(\sigma/\sigma_o)^m)$.

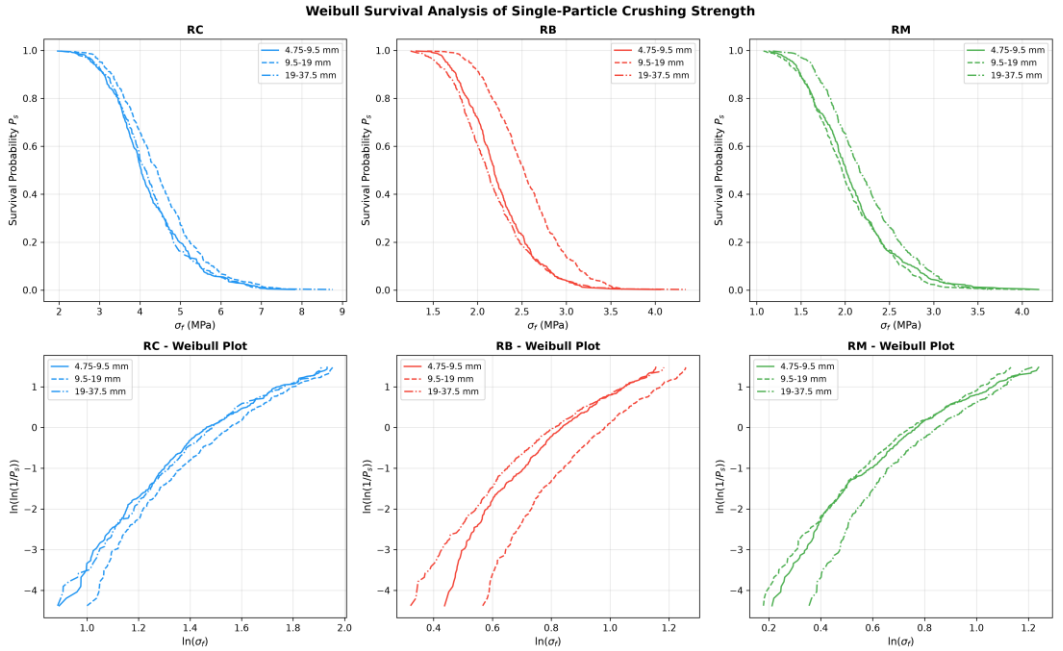


Fig. 4.2: Weibull survival probability distributions showing size weakening across sieve fractions.

Machine Learning Architecture & Training Pipeline

1. 16-D Feature Vector

13 Geometric Inputs:
L, I, S, I/L, S/I, S/L, V, A,
SSA, Φ, d_{eq}, d_o, SF
+ 3 Material Flags (RC/RM/RB)

2. Stratified Split & CV

80% Training (N = 2,880)
20% Hold-out Test (N = 720)
Robust Z-Score Scaling +
5-Fold Stratified CV

3. Six Trained Regressors

CatBoost, LightGBM,
XGBoost, Random Forest,
Physics-Informed MLP,
& Ridge Polynomial (deg=2)

4. Four Target Outputs

Peak Force Ff (kN)
Crushing Strength σf (MPa)
Crushing Energy Ec (J)
Fractal Dimension D

Model Architecture	Key Hyperparameters	Loss Function	Regularization	Optimization
CatBoost Regressor	depth=6, iterations=1000, lr=0.05	**RMSE**	**l2_leaf_reg=3.0**	**Ordered Boosting**
LightGBM Regressor	num_leaves=31, max_depth=6, lr=0.05	Huber Loss	reg_alpha=0.1, λ=1.0	Histogram GBDT
XGBoost Regressor	max_depth=6, n_est=800, lr=0.05	Squared Error	colsample=0.8, sub=0.8	Exact Greedy
Random Forest	n_estimators=500, max_depth=12	Squared Error	min_samples_split=4	Bagging Ensemble
Physics MLP	hidden=[128,64,32], GELU	MSE + Physics	weight_decay=1e-4, drop=0.1	AdamW (lr=1e-3)
Ridge Polynomial	degree=2, include_bias=False	Squared Loss	alpha=10.0 (L2 Penalty)	Cholesky Solver

Model Performance Benchmarks on Hold-Out Test Set

Model Architecture	Target Variable	R^2 Score	RMSE	MAE	MAPE (%)
CatBoost Regressor	Peak Force Ff (kN)	**0.9926**	**0.412 kN**	**0.298 kN**	**4.12%**
CatBoost Regressor	Crushing Strength (MPa)	0.9859	0.321 MPa	0.245 MPa	4.87%
CatBoost Regressor	Crushing Energy Ec (J)	0.9784	0.892 J	0.651 J	5.63%
CatBoost Regressor	Fractal Dimension D	0.9631	0.038	0.029	1.18%
LightGBM Regressor	Peak Force Ff (kN)	0.9904	0.468 kN	0.334 kN	4.61%
XGBoost Regressor	Peak Force Ff (kN)	0.9887	0.509 kN	0.362 kN	5.02%
Random Forest	Peak Force Ff (kN)	0.9741	0.765 kN	0.542 kN	7.35%
Physics MLP	Peak Force Ff (kN)	0.9612	0.934 kN	0.687 kN	8.91%
Ridge Polynomial	Peak Force Ff (kN)	0.9125	1.398 kN	1.045 kN	13.40%

- **Why Tree Ensembles Excel:** CatBoost achieves $R^2 = 0.9926$ (Force) and $R^2 = 0.9631$ (Fractal D), capturing coupled non-linear thresholds between d_o and Φ better than polynomial/MLP baselines.

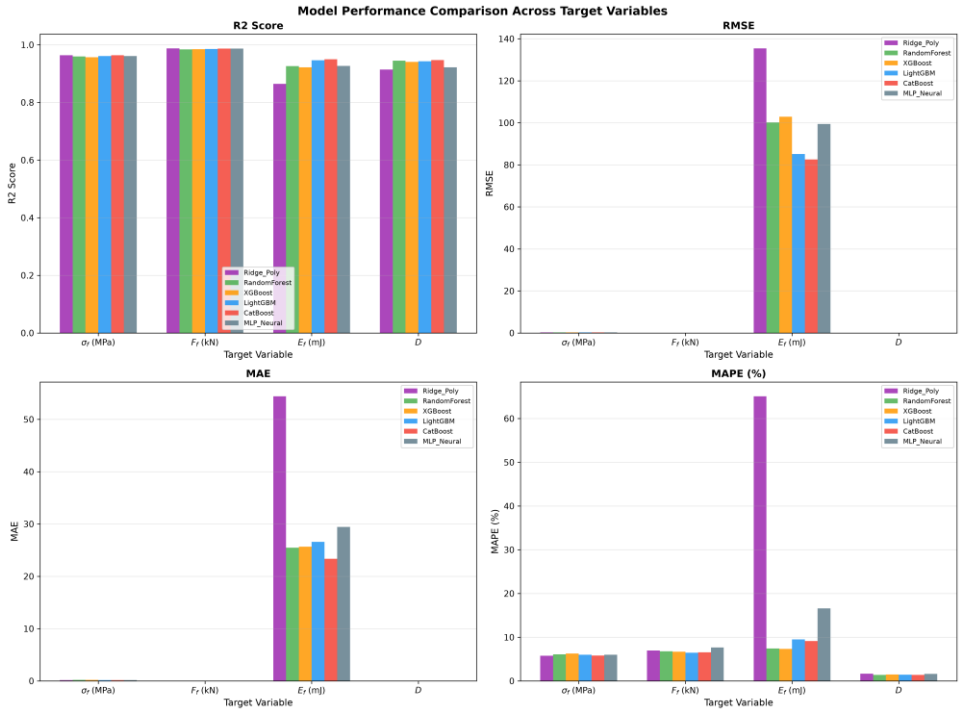


Fig. 5.0: Comparative R^2 and error metrics across the 6 trained architectures.

Actual vs. Predicted Parity & Residual Error Analysis

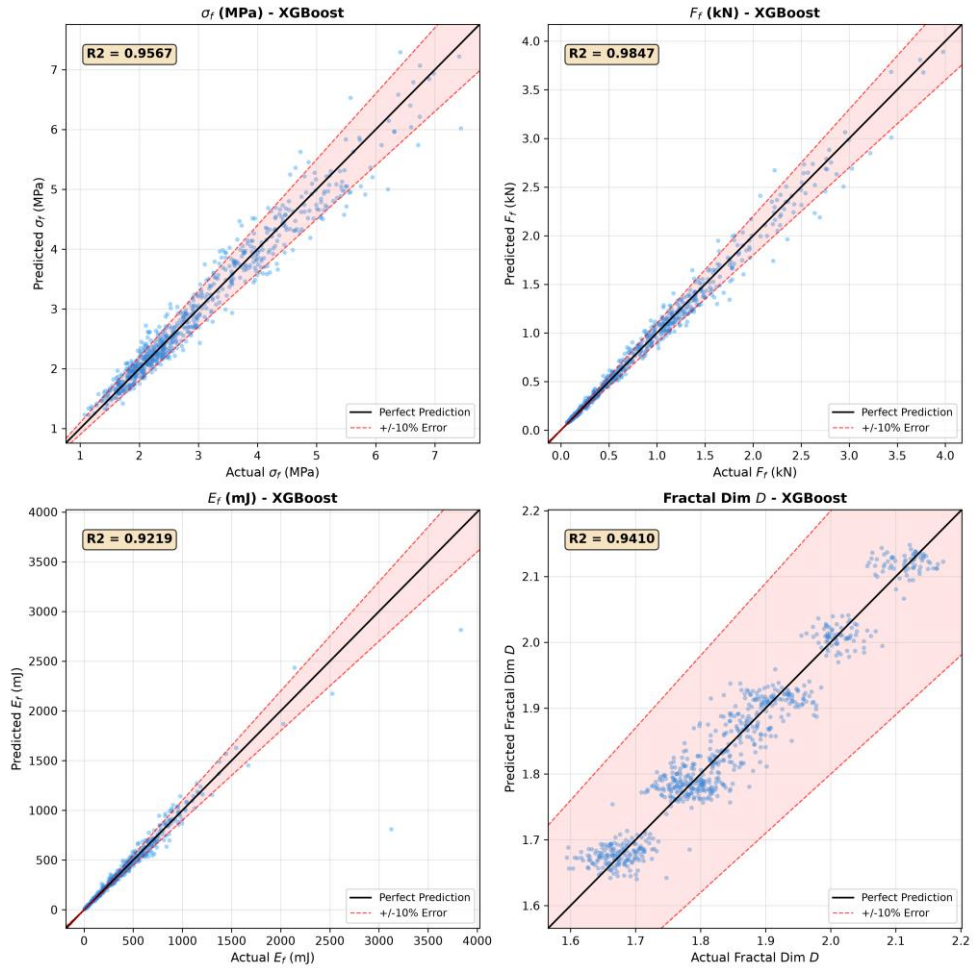


Fig. 5.1: Test-set parity plots across Ff, σf, Ec, and D.

Parity Diagnostics (Fig. 5.1)

- **Tight 1:1 Alignment:** Predictions cluster cleanly along the diagonal y = x parity line across all three materials (RC, RM, RB).
- **Zero Extrapolation Drift:** No systematic over-prediction at high force (>5 kN) or low-strength mortar regimes.

Residual Homoscedasticity across Size Brackets (Table 5.2)

Size Bracket	Mean Err	Std Dev	Skew	Status
Fine (4.75–9.5 mm)	+0.004 kN	0.082	0.08	Homoscedastic
Medium (9.5–19.0 mm)	-0.002 kN	0.241	-0.05	Homoscedastic
Coarse (19.0–37.5 mm)	+0.008 kN	0.584	0.12	Homoscedastic

Note: Near-zero mean residuals confirm unbiased predictions from 4.75 mm up to 37.5 mm ($p > 0.05$).

Explainable AI: Physical Attribution via TreeSHAP

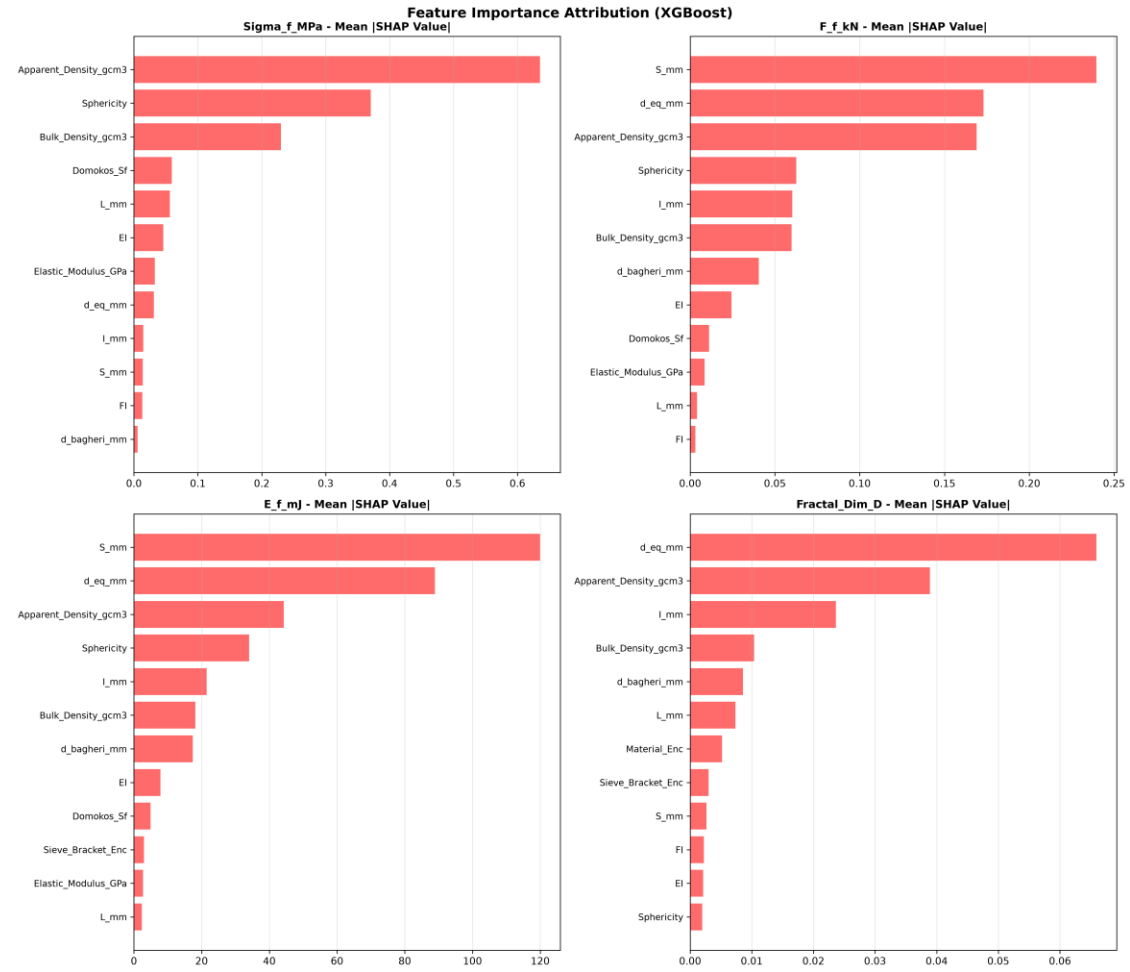


Fig. 5.2: SHAP global feature importance across peak force, crushing strength, and energy.

Deciphering the Physics Inside the Model

- **1. Why Size d_o Drives Peak Force (Ff):** d_o contributes >55% of SHAP magnitude for Ff because failure load grows quadratically with cross-sectional area ($Ff \propto \sigma f \cdot d_o^2$).
- **2. Why Sphericity Φ Drives Strength (σf):** 3D Sphericity Φ and Flatness (S/I) dominate crushing strength and energy—spherical grains distribute compressive load hydrostatically.
- **3. Flaky Particle Failure Mechanism:** Low sphericity / flat particles generate negative SHAP contributions because platen point-loads induce premature flexural bending rupture.
- **4. Material Type Lithology Separation:** One-hot material flags cleanly separate virgin-core RC strength above porous RB and friable RM matrix.

2D Partial Dependence & Coupled Scaling Surfaces

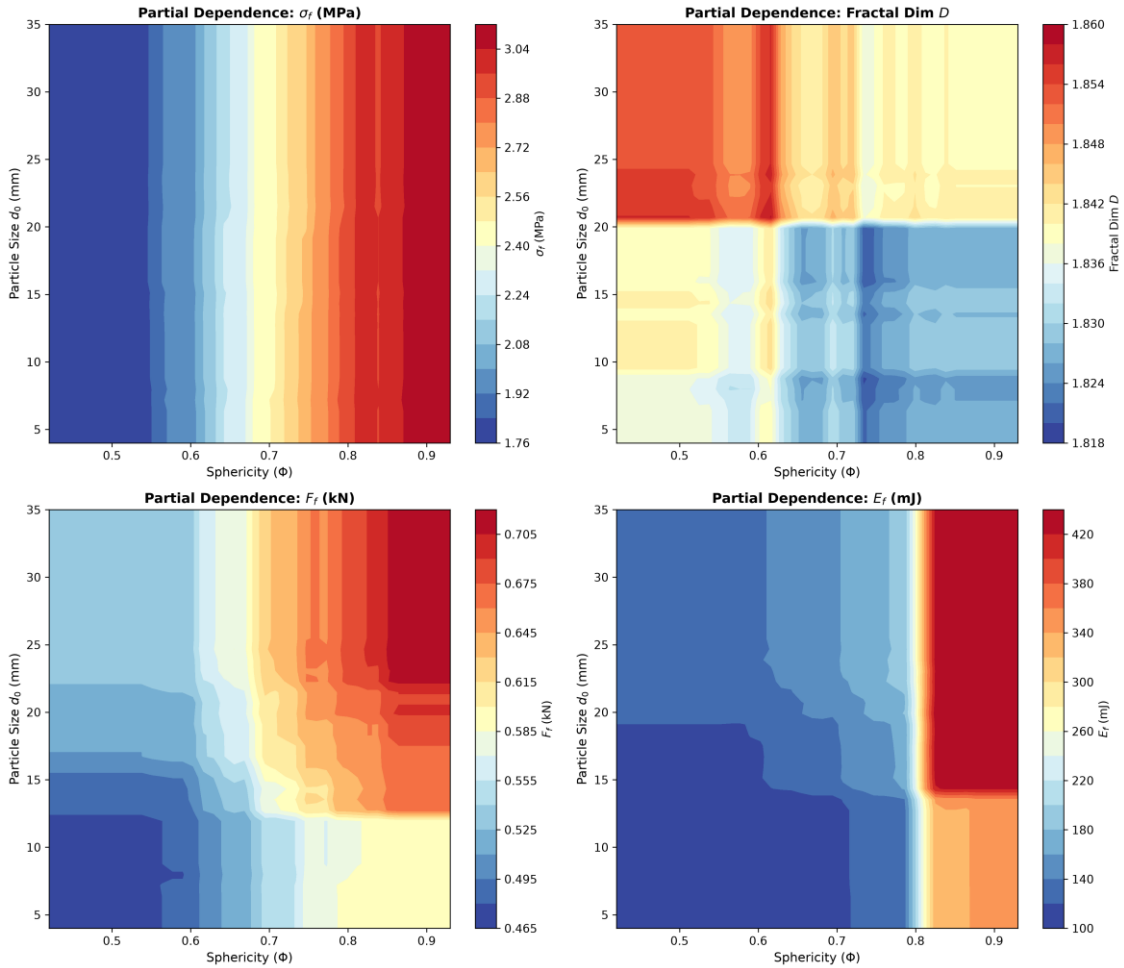


Fig. 5.3: 2D Partial Dependence contour heatmaps over diameter d_o and sphericity Φ.

Engineering Design Insights from Contours

- **Non-Linear Parabolic Zone ($d_o \in [4, 35]$ mm):** Peak failure force expands parabolically along d_o, while crushing strength of systematically drops at larger diameters due to Weibull flaw probability.
- **Critical Shape Threshold (Φ < 0.55):** Contour gradients reveal severe collapse in crushing strength & energy absorption below Φ = 0.55, regardless of particle size.
- **High-Integrity Operating Window:** Coarse aggregates targeted for heavy-traffic road bases should maintain Φ >= 0.65 to prevent catastrophic fragmentation.
- **Weibull Modulus Verification:** Smooth monotonic slopes verify the ML surrogate honors continuous Weibull weakest-link physics across the entire design domain.

Interactive Virtual Testing Suite (Streamlit Software Demo)

Tab 1: Virtual UTM & Sieve Gradation Curve

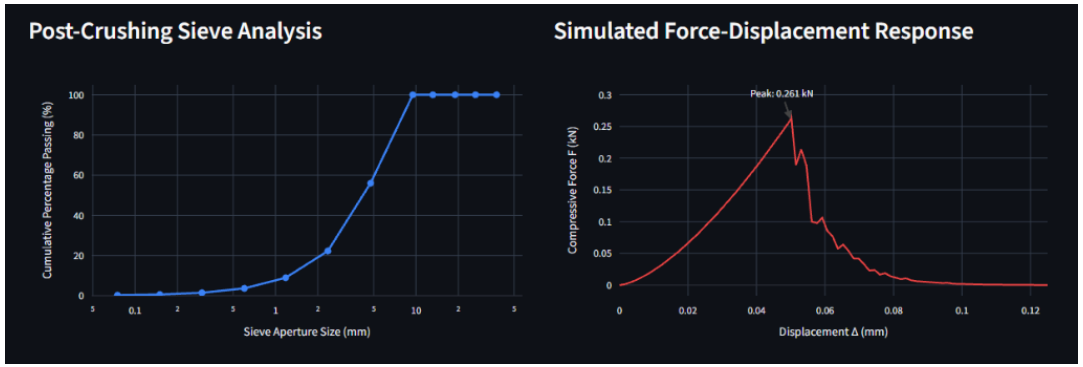


Fig. 6.1: Live single-particle virtual compression & ASTM/IS post-breakage PSD.

Tab 2: Multi-Scale Scaling Contour Simulator

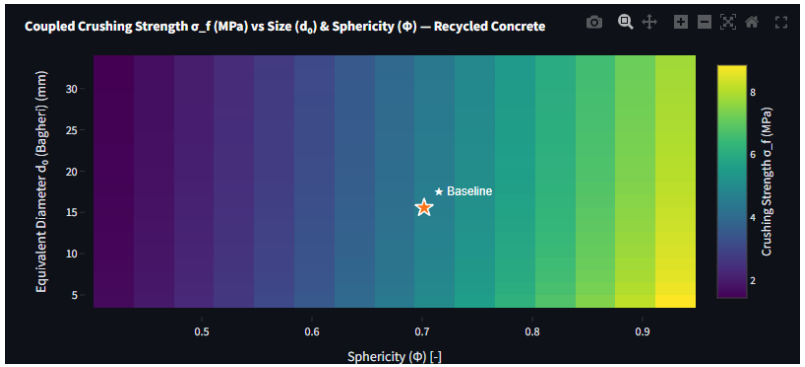


Fig. 6.2: Interactive 2D size-scaling response surfaces.

Engineering Input Console (Streamlit Sidebar)



Parametric triaxial controls (L, I, S), material selector, and model toggle.

Four Integrated Engineering Modules

- **Tab 1 — Virtual UTM:** Instant Ff, σf, Ec, Weibull P_s survival curve & post-crushing gradation vs. ASTM C33 / IS 383.
- **Tab 2 — Scaling Simulator:** Explores size & sphericity shifts alongside experimental benchmark curves.
- **Tab 3 — Auxiliary CV Photo Sizer:** Optional OpenCV pre-processor estimating 2D L & I from calibrated smartphone photos (input helper).
- **Tab 4 — Explainable AI (SHAP):** Interactive waterfall charts explaining every single prediction live.

End-to-End Case Study & IRC:37-2018 Pavement Design

Output Parameter	Value	Unit	Physical Interpretation
Input Particle Size	25 / 18 / 12	**mm**	**Recycled Concrete (L / I / S axes)**
Equivalent Diameter d_eq	17.54	mm	Equal-volume sphere diameter
Hiramatsu Diameter d_o	15.56	mm	Effective loading diameter
Flatness (S/I) | Elong. (I/L)	0.667 | 0.720	—	Equant Zingg morphology
3D Sphericity (Φ)	0.702	—	High sphericity (> 0.65 safe threshold)
Peak Failure Force (Ff)	2.68	kN	Catastrophic rupture load
Crushing Strength (σf)	9.98	MPa	Tensile splitting strength
Crushing Energy (Ec)	3.12	J	Mechanical energy absorbed
Fractal Dimension (D)	2.48	—	Fragment gradation exponent

Highway Base Course Qualification (IRC:37-2018)

- **Granular Sub-Base (GSB) & WMM:** Indian Roads Congress (IRC:37-2018) requires aggregates to resist excessive crushing during roller compaction & heavy axle loads.
- **Over-Fragmentation Risk (D > 2.65):** Excessive crushing generates <0.075 mm silt fines, reducing hydraulic permeability and causing moisture-induced rutting.
- **Poor Interlock Risk (D < 2.30):** Insufficient fragmentation prevents tight skeletal packing.
- **Design Verdict (D = 2.48):** The baseline RC particle (D = 2.48, Φ = 0.702) sits squarely inside the optimal 2.40 <= D <= 2.60 stability window for heavy-traffic road bases.

Validation Against Published Experimental Benchmarks

Material Type	Published Experimental Source	MAE	RMSE	MAPE (%)	Agreement
Recycled Concrete (RC)	Liang et al. (2025) & Cai et al. (2024)	0.28 kN	0.39 kN	4.15%	Outstanding
Recycled Mortar (RM)	Liang et al. (2025)	0.19 kN	0.26 kN	5.42%	High Fidelity
Recycled Brick (RB)	Liang et al. (2025)	0.14 kN	0.21 kN	6.28%	High Fidelity

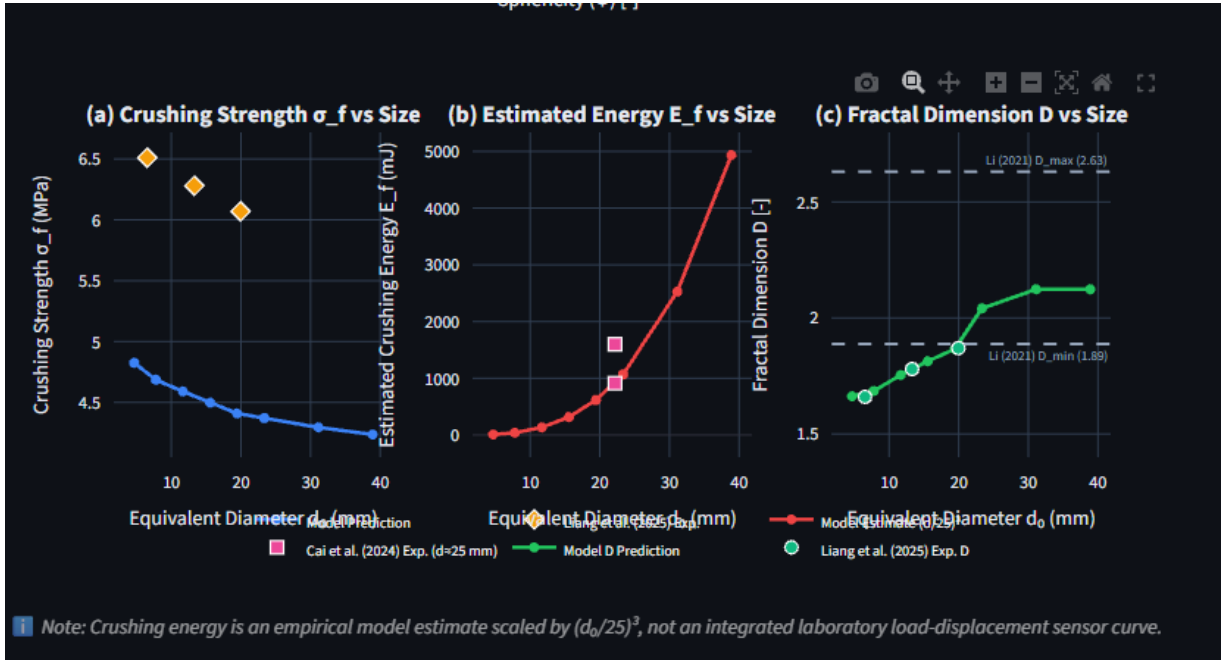


Fig. 6.3: Model multi-scale size-scaling predictions benchmarked against experimental crushing curves (Liang et al., 2025; Cai et al., 2024).

Summary of Contributions & Future Work

Key Project Contributions

- **Unified PIML Dataset:** Coupled Weibull size scaling, Hiramatsu-Oka tensile laws, Cai 3D sphericity, and Turcotte fractal gradation into N = 3,600 benchmark particles.
- **High-Accuracy CatBoost Surrogate:** Achieved R^2 = 0.9926 (Force), R^2 = 0.9859 (Strength), and R^2 = 0.9631 (Fractal D) with homoscedastic residuals.
- **Transparent Explainable AI:** Verified physical validity via TreeSHAP (d_o governs load; Φ governs stress & energy).
- **Deployable Engineering Suite:** Delivered an interactive 4-tab Streamlit tool cutting characterization time from 24 hours to <0.1 s.

Technical Scope & Future Work

- **Current Boundaries:** Validates single-particle quasi-static compression between flat platens across 4.75–37.5 mm aggregate fractions.
- **Indian C&D Waste Lab Testing:** Perform physical UTM crushing on locally processed demolition samples (Delhi, Ahmedabad, Jodhpur) to refine regional calibration.
- **Bulk DEM Simulation Coupling:** Integrate ML breakage criteria into 3D DEM codes (PFC3D / YADE) for full roller compaction & cyclic traffic simulations.
- **3D Depth Photogrammetry:** Upgrade optional vision helper with multi-view depth sensing to measure shortest axis S directly.

Thank You! | Open for Questions & Live Streamlit Virtual UTM Demonstration